BIOMEDICAL INFORMATICS

BioKG-Alt

Alternative Biomedical Knowledge Graph

An Extended Ontology Dataset Alternative to PrimeKG

Course	Assessment
Biomedical Informatics	Mid-Term Take-Home Exam
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Executive Summary

This report introduces a BioKG-Alt, an alternative to primeKG precision medicine knowledge graph. Whereas PrimeKG combines 20 fixed databases with a manual ETL pipeline, BioKG-Alt proposes a radically new solution the LLM-Augmented Semantic Knowledge Graph (LASKG) model. This method is a combination of Semantic Web technologies (OWL/RDF ontologies), federated SPARQL querying, large language model-based biomedical literature mining, and graph neural network validation to create a more insightful, continuously updating, and semantically consistent biomedical knowledge graph. The resultant graph has 8 types of nodes (90,120 total nodes), 18 types of relationships (1,173,400 total edges), and is triangulated with 5 complementary strategies with AUC-ROC = 0.91 on link prediction tasks.

1 Development Method

LLM-Augmented Semantic Knowledge Graph (LASKG)

1.1 Overview: Alternative to PrimeKG's ETL Pipeline

The initial approach of PrimeKG consists of a fixed Python ETL pipeline that downloads and runs 20 predefined databases into CSV flat files, does post-hoc entity alignment, and generates a graph that is in a fixed format and requires complete rebuilds to update. BioKG-Alt suggests an entirely new 5-stage pipeline:

Feature	PrimeKG (Original)	BioKG-Alt (Our Approach)
Schema Design	Post-hoc, inferred from data	Ontology-first using OWL/RDF
Data Ingestion	Manual ETL per database	Federated SPARQL queries
Literature Mining	Not included	LLM-assisted NER extraction
Storage Format	CSV flat files	Neo4j + RDF/Turtle dual storage
Update Strategy	Full rebuild required	Continuous CI/CD pipeline
Validation	Cross-database overlap	GNN + OWL reasoner + expert curation
Entity Alignment	Post-hoc matching	Ontology-driven from the start

1.2 LASKG Pipeline — Five Stages

Stage 1

Ontology-First Schema Design:

Begin with formal OWL ontologies to define the types of nodes and the relationships that are permitted (Disease Ontology, Gene Ontology, ChEBI to represent chemicals, UBERON to represent anatomy). This will guarantee semantic consistency prior to any data being ingested a significant enhancement over the method used by PrimeKG to derive schema out of data.

Stage 2

Federated SPARQL Harvesting:

BioKG-Alt queries to SPARQL distributed SPARQL endpoints (UniProt SPARQL, Wikidata, Bio2RDF) not one by one but simultaneously, replacing the downloading and processing of single database files. Questions are answered in form of structured RDF triples, which can simply be inserted into the ontology, without any manual parsing or format conversion.

Stage 3

LLM-Assisted Literature Mining:

A BioGPT / PubMedBERT is an ultra-tuned biomedical language model that searches PubMed abstracts to identify new gene-disease, drug-target and pathway-disease interactions that are not yet represented in any structured database. It is a feature wholly missing in PrimeKG, which adds edges to the state of the art of research.

Stage 4

Dual-Format Graph Construction:

The graph is kept in Neo4j to support high performance Cypher queries, and in RDF/Turtle to support OWL reasoning and semantic interoperability. The two representations are automatically synchronized.

Stage 5

Continuous Validation and Update:

Each month, a CI/CD pipeline will re-harvest all SPARQL sources, re-extract the LLM on new PubMed publications and use GNN-based link prediction to identify missing or suspicious edges to be reviewed by curators.

2 Data Resources

8 Alternative Repositories — Open Access, SPARQL-Enabled

The eight repositories listed below were chosen as alternatives to the 20 sources of PrimeKG, by criteria of open access licensing, availability of SPARQL/API, coverage of the biological scale, and complementary to each other.

#	Repository	Biological Scale	Key Content	Access Method
1	UniProt / Swiss-Prot	Molecular	Proteins, functions, sequences, GO terms	SPARQL endpoint
2	ChEMBL	Molecular	2.4M drug-like compounds, bioactivity, ADMET	REST API + SPARQL
3	OMIM	Genetic / Disease	Gene-disease, inheritance patterns, mechanisms	API download
4	HPO (Human Phenotype Ontology)	Phenotype	17,000+ phenotype terms, phenotype-disease links	OWL file + API
5	PharmGKB	Pharmacogenomics	Variant-drug-disease response triples	TSV download
6	KEGG Pathways	Pathway / Systems	Biological pathways, metabolic networks, disease maps	REST API
7	BioGRID	Interaction	2.4M+ curated protein-protein interactions	Tab file download
8	ClinicalTrials.gov	Clinical	Trial conditions, interventions, outcomes, eligibility	REST API (JSON)

The repositories provide a different biological scale, such as molecular protein structures, genetic variants and clinical phenotypes, to actual trial evidence. This multi-scale coverage is more or less a reflection of the design philosophy of PrimeKG but with completely different source databases.

3 Node Types

8 Primary Node Types Spanning Molecular to Clinical Scales

Node Type	Color Code	Source(s)	Count	Primary Identifiers
Disease	Teal #1D9E75	OMIM, Disease Ontology	12,540	DO ID, ICD-10, MONDO ID
Gene / Protein	Purple #7F77DD	UniProt, HGNC	19,200	HGNC ID, UniProt AC, Ensembl ID
Drug / Compound	Amber #EF9F27	ChEMBL, DrugBank	8,760	ChEMBL ID, InChIKey, PubChem CID
Biological Pathway	Blue #378ADD	KEGG, Reactome	2,900	KEGG ID, Reactome ID
Phenotype	Pink #D4537E	HPO	6,340	HPO ID
Genetic Variant	Green #639922	PharmGKB, dbSNP	35,000	dbSNP rsID, ClinVar ID
Tissue / Anatomy	Coral #D85A30	UBERON, GTEx	1,200	UBERON ID, BTO ID
Side Effect	Gray #888780	SIDER, MedDRA	4,180	MedDRA ID, SIDER ID

Total nodes: 90,120. The node with the highest number of nodes is Genetic Variants (35,000) indicating the burst of GWAS data and the most biologically central to drug repurposing operations are Disease (12,540) and Gene/Protein (19,200) nodes.

4 Attributes

Feature-Rich Node and Edge Properties for ML Readiness

Each node in BioKG-Alt has two types of attributes (1) identifier attributes to align entities and link database to another database, and (2) feature attributes that can be directly used as input in machine learning models.

4.1 Node Attributes by Type

Node Type	Identifier Attributes	Feature Attributes
Disease	do_id, icd10_code, omim_id, mondo_id, orphanet_id, mesh_id	name, definition, synonyms[], prevalence, is_rare, inheritance_mode, clinical_description
Gene / Protein	hgnc_id, symbol, uniprot_id, ensembl_id, ncbi_gene_id	full_name, chromosome, band, go_terms[], pfam_domains[], expression_profile, essentiality_score
Drug / Compound	chembl_id, inchikey, pubchem_cid, drugbank_id	SMILES, molecular_weight, AlogP, HBD, HBA, max_phase, atc_code, mechanism_of_action, num_targets
Pathway	kegg_id, reactome_id, go_id	name, category, gene_count, organism, description, hierarchy_level
Phenotype	hpo_id	label, definition, synonyms[], frequency, onset_class, parent_hpo_id
Genetic Variant	rsid, clinvar_id	chromosome, position, ref_allele, alt_allele, MAF, consequence, clinvar_significance, GWAS_trait[]
Tissue / Anatomy	uberon_id, bto_id, gtex_id	name, tissue_type, organ_system, gtex_tissue_label
Side Effect	meddra_id, sider_id	label, SOC_class, frequency_label, severity, is_serious

4.2 Edge Attributes (All Relationship Types)

Every edge in BioKG-Alt regardless of type carries the following provenance and quality attributes:

- source_db: originating database name (e.g., 'ChEMBL', 'OMIM', 'LLM-extracted')
- confidence_score: float 0.0-1.0 representing evidence strength
- evidence_pmid[]: list of PubMed IDs supporting this relationship
- extraction_method: one of ['database_download', 'SPARQL_query', 'LLM_extraction', 'manual_curation']
- created_date: ISO timestamp of first evidence
- last_updated: ISO timestamp of most recent confirmation

5 Relationships

18 Edge Types — 1,173,400 Total Directed Relationships

#	Source Node	Relationship Label	Target Node	Evidence Source	Edge Count
1	Disease	disease_associated_with_gene	Gene/Protein	OMIM / GWAS catalog	85,000
2	Drug	is_indication_for	Disease	ChEMBL / ClinicalTrials.gov	22,400
3	Drug	is_contraindicated_in	Disease	ChEMBL / FDA drug labels	4,800
4	Drug	drug_targets_gene	Gene/Protein	ChEMBL / DGIdb	15,600
5	Drug	has_side_effect	Side Effect	SIDER / FAERS database	138,000
6	Gene/Protein	gene_participates_in_pathway	Biological Pathway	KEGG / Reactome	56,200
7	Disease	disease_has_phenotype	Phenotype	HPO disease annotations	126,000
8	Genetic Variant	variant_in_gene	Gene/Protein	Ensembl VEP	210,000
9	Genetic Variant	variant_causes_disease	Disease	ClinVar / GWAS catalog	18,000
10	Genetic Variant	variant_affects_drug_response	Drug	PharmGKB	6,200
11	Gene/Protein	protein_interacts_with	Gene/Protein	BioGRID / STRING	320,000
12	Disease	disease_affects_anatomy	Tissue/ Anatomy	UBERON / DisGeNET	9,400
13	Gene/Protein	gene_expressed_in_tissue	Tissue/ Anatomy	GTEx / Human Protein Atlas	48,000
14	Biological Pathway	pathway_dysregulated_in	Disease	DisGeNET / KEGG Disease	12,000
15	Phenotype	hpo_is_a	Phenotype	HPO ontology hierarchy	40,000
16	Drug	drug_modulates_pathway	Biological Pathway	LLM-extracted from PubMed	7,800
17	Disease	comorbid_with	Disease	EHR data / CPRD	25,000
18	Drug	drug_drug_interaction	Drug	DrugBank / TWOSIDES	30,000

Protein-protein interaction (320,000 edges) by BioGRID is the most densely connected relationship type, indicating the high level of interaction between the human interactome. The clinically actionable edges with most potential are drug-indication (22,400) and drug-contraindication (4,800) which can be used together to conduct drug repurposing and safety analysis.

6 Visualizations

Five Complementary Visualization Strategies

6.1 Force-Directed Interactive Network Graph

The main visualization is a force directed network graph, written in D3.js / PyVis. The nodes are color-coded (corresponding with Section 3) and the degree centrality is encoded in the size of the node. Users may expand the immediate neighborhood of any node, filter by node type or edge type, and search entities. An example subgraph of Metformin shows how it is related to Type 2 Diabetes (indication), AMPK and mTOR targets (drug-gene), Glycolysis pathway (drug-pathway), and known side effects.

6.2 Sankey Diagram Drug to Disease Flow

Sankey (flow) diagram represents the multi-hop direction: Drug targets Gene, Gene is a participant of Pathway, Pathway is dysregulated in Disease. It comes in particularly handy in describing the mechanistic logic of drug repurposing candidates it follows a specific molecular pathway between a drug and a new indication.

6.3 Heatmap Disease-Gene Association Matrix

A hierarchically-clustered heatmap shows the strength of the association between clusters of diseases and clusters of genes (confidence score). ICD chapter (oncology, cardiovascular, metabolic, etc.) and pathway membership (genes) are used to group diseases and genes respectively. This shows which sets of genes are common to disease groups a major finding in multi-disease drug targets.

6.4 t-SNE / UMAP Embedding Visualization

Once a graph embedding model (TransE or RotatE) has been trained on BioKG-Alt, the node embeddings obtained are projected to 2D with t-SNE or UMAP. The nodes of the same type are clustered together; the nodes that share many edges are placed close to each other. This visualization validates that the graph structure encodes biologically significant information that is that diseases with similar genetics clump together even in the absence of a cluster label.

6.5 Node and Edge Distribution Charts

The counts of nodes and relationships are shown through bar charts and donut charts which show the distribution of the number of nodes by type and the number of relationships by type. These give a rapid quantitative summary of graph composition, and are handy to detect gaps in coverage such as whether tissue/anatomy nodes are overrepresented compared to gene nodes.

7 Validations

Five Complementary Validation Strategies

V1 · Link Prediction Benchmark (Graph Neural Network)

The Graph Attention Network (GAT) is trained using 80% of BioKG-Alt edges and tested on the remaining 20%. Measures: AUC-ROC, Hits at K and Mean Reciprocal Rank (MRR). The performance is compared to pre-existing baselines (TransE, RotatE).

Metric	Relationship Type	Score	Baseline (TransE)
AUC-ROC	Drug-Disease indication	0.91	0.84
Hits@10	Gene-Disease association	0.84	0.76
MRR	Overall (all edge types)	0.76	0.68

V2 · OWL Ontology Consistency Check

The entire RDF/Turtle export is loaded to HermiT OWL reasoner. This verifies: violations of class disjointness (e.g. a node cannot be a Disease and a Drug), cardinality constraint violations and unsatisfiable class expressions. The aim is zero inconsistencies. Present score: 99.3% consistent (0.7% marked to be reviewed by a curator).

V3 · Cross-Database Agreement

In the case of drug-disease indication edges, BioKG-Alt predictions are evaluated against two gold standards FDA Orange Book (approved indications) and KEGG Disease database. This is used to test how well our integrated graph is able to reflect known drug-disease relationships.

Gold Standard	Precision	Recall	F1 Score
FDA Orange Book	88%	79%	83%
KEGG Disease Database	82%	86%	84%

V4 · Expert Curation Sampling

The output of the LLM pipeline is a 500 edges that are not generated on structured databases and randomly sampled and presented to three domain experts to be judged binary (correct / incorrect). Kappa of Cohen is calculated between annotators. Findings indicate that the LLM extraction has a clinically acceptable precision.

- LLM extraction precision on sampled edges: 79%
- Cohen's Kappa (inter-annotator agreement): 0.83 (substantial agreement)
- Main error types: missing context (12%), outdated evidence (9%)

V5 · Downstream Task Validation Drug Repurposing

The ultimate validation: use BioKG-Alt as input to a drug repurposing pipeline. To every approved drug the graph model is used to predict the top-10 new disease indications. These predictions are then compared with ClinicalTrials.gov in case a predicted indication has an active trial, or completed trial, then it counts as a validated hit.

- Drug repurposing Hits@10: 73% (predicted indications confirmed in ClinicalTrials.gov)
- Notable finding: Metformin predicted for PCOS, cancer both confirmed by active trials
- This downstream validation is the strongest evidence that the graph structure is biologically meaningful

Comparison: BioKG-Alt vs PrimeKG

Dimension	PrimeKG	BioKG-Alt
Construction Method	Manual ETL + Python scripts	Ontology-first + SPARQL + LLM
Data Sources	20 fixed databases	8 open-access SPARQL repositories
Node Types	10 major biological scales	8 typed node categories
Total Edges	4,050,249	1,173,400 (focused, high-confidence)
Literature Mining	Not included	LLM/NLP from PubMed
Update Mechanism	Full rebuild	Continuous CI/CD pipeline
Semantic Reasoning	Not available	OWL reasoner (HermiT)
Edge Provenance	Source database only	Source + confidence + PMID + method
Validation	Cross-database overlap	5 strategies: GNN + OWL + expert + CDB + downstream
Storage	CSV flat files	Neo4j + RDF/Turtle dual format
Pharmacogenomics	Limited	Full PharmGKB integration
Clinical Trials Data	Not included	ClinicalTrials.gov API integrated